Transfer task: defend and revise

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Defend the prediction with evidence and make a visible, justified revision.

SUCCESS CRITERIA - I can...

1. Graph the central prediction and coordinate bounds.
2. Defend a claim using calculations and model limits.
3. Document a specific revision with evidence.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “evidence” and “revision” to explain your model or result. Sentence starters are optional.

evidence: observations or measurements that support an idea

revision: a change made after checking evidence or reasoning

prediction: what a model says may happen next

01 Bring back an earlier idea

For x_i=3±1 m, v=5±0.5 m/s and exact t=7 s, find the central final position and conservative interval.
Name a specific check of your method.

Word help: conservative bound = a limit chosen to include every allowed case

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $x_f = x_i + v_x t$; $y_f = y_i + v_y t$.
Use your Day 21 data and consistent units. Defend the endpoint and assumptions. A bounded range is not a probability.

Worked reasoning: A cart model predicts 12 m after 4 s; the observation is 10±0.5 m. Since 12 is outside 9.5-10.5 m, the model needs checking. The mismatch alone does not identify the cause.

Your turn: Mark the observed interval and prediction on a number line.
Name one model assumption to recheck.

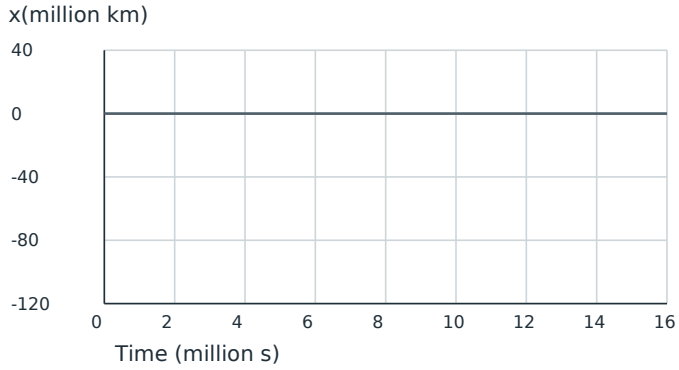
Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: ① ② ③

03 Graph central and bounded positions

Use Day 21 Honors data. Graph central, lower and upper x(t) from 0 to 16 million seconds.
Use additional graph paper if needed.
Mark end point values and explain the changing width.

Sentence starter: Under the assumption …, the evidence … supports … because … .



Marzano check - circle: ① ② ③

04 Defend the public claim

Write an accessible paragraph with central end point, coordinate intervals, and the meaning of those bounds. Distinguish measurement uncertainty from constant-velocity model error.

Sentence starter: The step I would change is … . The evidence is … .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Peer defense and revision

Explain your method to a partner. Have them ask about units, axes, bounds or assumptions.
Record one question and one evidence-based revision. Keep the original visible. If alone, label a self-review.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

A 5 kg object hangs still from a vertical string.
Use g=9.8 N/kg.
Name all forces on it with sources, sizes and directions. Does object-on-string belong on this diagram?

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: 1 2 3

07 Exit check

Identify evidence for Science, Reasoning, Communication, and Transfer to asteroid from your finished work.
Name your most important revision and one next-unit question.

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A class fits v=2.5 m/s using x=10 m at 4 s. It then measures x=13.0±0.2 m at 6 s. Evaluate the new evidence and propose a two-interval model consistent with the central measurements.

Word help: evaluate = check a claim against evidence and conditions

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3